Lista -> 2 G.A   Gabriela Vinha Feitosa

1- a) $\begin{vmatrix} 1 & 2 \\ -4 & 3 \end{vmatrix}$

$Det = 2 \cdot C_{12} + 3 \cdot C_{22}$

$Det = 2 \cdot (-1)^{1+2} \cdot D_{12} + 3 \cdot (-1)^{2+2} \cdot D_{22}$

$Det = 2 \cdot -D_{12} + 3 \cdot D_{22}$

$Det = 2 \cdot -(-4) + 3 \cdot 1$

$Det = 8 + 3 \Rightarrow Det\ 11$

b)- $\begin{vmatrix} \sqrt{2} & 3\sqrt{6} \\ 2 & \sqrt{3} \end{vmatrix}$

$Det = 3\sqrt{6} \cdot C_{12} + 3\sqrt{3} \cdot C_{22}$

$Det = 3\sqrt{6} \cdot (-1)^{1+2} \cdot D_{12} + \sqrt{3} \cdot (-1)^{2+2} \cdot D_{22}$

$Det = 3\sqrt{6} \cdot -D_{12} + \sqrt{3} \cdot D_{22}$

$Det = 3\sqrt{6} \cdot -2 + \sqrt{3} \cdot \sqrt{2}$

$Det = -6\sqrt{6} + \sqrt{6}$

$Det = -5\sqrt{6}$

c) $\begin{vmatrix} \pi & 0 & 2 \\ 5 & -1 & 1 \\ 1 & 0 & 0 \end{vmatrix}$

$Det = 1 \cdot C_{31} + 0 \cdot C_{32} + 0 \cdot C_{33}$

$Det = 1 \cdot (-1)^{3+1} \cdot D_{31}$

$Det = + D_{31} \Rightarrow \begin{vmatrix} 0 & 2 \\ -1 & 1 \end{vmatrix} = 2$

$Det = 2$

d) $\begin{vmatrix} -2 & 1 & -1 \\ 1 & 5 & 4 \\ -3 & 4 & 2 \end{vmatrix}$

$Det = -2 \cdot C_{11} + 1 \cdot C_{12} + (-1) \cdot C_{13}$

$Det = -2 \cdot (-1)^{1+1} \cdot D_{11} + 1 \cdot (-1)^{1+2} \cdot D_{12} + (-1) \cdot (-1)^{1+3} \cdot D_{13}$

$Det = -2 \cdot \begin{vmatrix} 5 & 4 \\ 4 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 4 \\ -3 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 5 \\ -3 & 4 \end{vmatrix}$

$Det = -2 + (-(6+10)) - (12+2) - (15+4)$

$Det = -2 \cdot (-6) - 14 - 19$

$Det = 12 - 14 - 19 = -21$

e) $\begin{vmatrix} 0 & 2 & 0 \\ 1 & 3 & 5 \\ 2 & -1 & 2 \end{vmatrix}$

$Det = 0 \cdot C_{11} + 2 \cdot C_{12} + 0 \cdot C_{13}$

$Det = 2 \cdot (-1)^{1+2} \cdot D_{12}$

$Det = -2 \cdot \begin{vmatrix} 1 & 5 \\ 2 & 2 \end{vmatrix}$

$Det = -2 \cdot (-8) = 16$

$Det = 16$

f) $\begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{vmatrix}$

$Det = 3 \cdot C_{11} + 0 \cdot C_{21} + 0 \cdot C_{31} + 0 \cdot C_{41}$

$Det = 3 \cdot (-1)^{1+1} \cdot D_{11}$

$Det = 3 \cdot \begin{vmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix}$

$Det = 3 \cdot (-2)$

$Det = -6$

g) $\begin{vmatrix} 1 & 0 & 0 & 0 & 0 \\ 5 & 1 & 2 & 5 & 3 \\ 7 & 2 & \sqrt{5} & 0 & 0 \\ 10 & -3 & 6 & 1 & 0 \\ -2 & -3 & 0 & 0 & 0 \end{vmatrix}$

$Det = 1 \cdot C_{11} + 0 \cdot C_{12} + 0 \cdot C_{13} + 0 \cdot C_{14} + 0 \cdot C_{15}$

$Det = 1 \cdot (-1)^{1+1} \cdot D_{11}$

$Det = 1 \cdot \begin{vmatrix} 1 & 2 & 5 & 3 \\ 2 & \sqrt{5} & 0 & 0 \\ -3 & 6 & 1 & 0 \\ -3 & 0 & 0 & 0 \end{vmatrix}$

$Det = 1 \cdot 3 \cdot (-1)^{1+4} \cdot D'_{14}$

$Det = 1 \cdot (-3) \cdot \begin{vmatrix} 2 & \sqrt{5} & 0 \\ -3 & 6 & 1 \\ -3 & 0 & 0 \end{vmatrix}$

$Det = -3 \cdot -3\sqrt{5}$

$Det = 9\sqrt{5}$

h) $\begin{vmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -2 & 0 & 0 & 0 \end{vmatrix}$

$Det = 0 \cdot C_{41} + 0 \cdot C_{42} + 0 \cdot C_{43} + 1 \cdot C_{44} + 0 \cdot C_{45}$

$Det = C_{44}$

$Det = (-1)^{4+4} \cdot D_{44}$

$Det = 1 \cdot \begin{vmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & -2 & 0 & 0 \end{vmatrix}$

$Det = 1 \cdot 3 \cdot (-1)^{1+1} \begin{vmatrix} 0 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 0 \end{vmatrix}$

$Det = 1 \cdot 3 \cdot (-8)$

$Det = -24$

2-a) $Det(A+B) \Rightarrow A = \begin{pmatrix} 3 & -5 & 7 \\ 4 & 2 & 8 \\ 1 & -9 & 6 \end{pmatrix} + B = \begin{pmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -4 \end{pmatrix}$

$Det(A+B) = \begin{vmatrix} 7 & -2 & 14 \\ 3 & 2 & 10 \\ 4 & -8 & 2 \end{vmatrix}$

$-112 + 560 + 12 + 28 - 336 - 80$

$Det(A+B) = 72$

b) $Det(AB) \Rightarrow$ sabendo que $Det(A \cdot B) = Det A \cdot Det B$

$Det(A \cdot B) = \begin{vmatrix} 3 & -5 & 7 \\ 4 & 2 & 8 \\ 1 & -9 & 6 \end{vmatrix} \cdot \begin{vmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -4 \end{vmatrix}$

$Det(A \cdot B) = (36 - 252 - 40 - 14 + 216 + 120) \cdot (-7 + 18 + 8 - 12)$

$Det(A \cdot B) = 66 \cdot 7$

$Det(AB) = 462$

c) $det(B^t A^t)$; sabendo que $Det(B^t \cdot A^t) = Det(B^t) \cdot Det(A^t)$ e também $Det(A^t) = -Det(A)$ e $Det(B^t) = -Det(B)$

$Det(B^t A^t) = -7 \cdot -66$

$Det(B^t A^t) = 462$

d) $det(3A - 2C + B)$

$3 \cdot \begin{pmatrix} 3 & -5 & 7 \\ 4 & 2 & 8 \\ 1 & -9 & 6 \end{pmatrix} - 2 \begin{pmatrix} 2 & 3 & -1 \\ 6 & 9 & -2 \\ 8 & 12 & -3 \end{pmatrix} + \begin{pmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -4 \end{pmatrix}$

$\begin{pmatrix} 9 & -15 & 21 \\ 12 & 6 & 24 \\ 3 & -27 & 18 \end{pmatrix} - \begin{pmatrix} 4 & 6 & -2 \\ 12 & 18 & -4 \\ 16 & 24 & -6 \end{pmatrix} + \begin{pmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -4 \end{pmatrix}$

$\begin{pmatrix} 9 & -18 & 30 \\ -1 & -12 & 30 \\ -10 & -50 & 20 \end{pmatrix} \Rightarrow Det$

FORONI

$$Det = \begin{vmatrix} 9 & -18 & 30 \\ -1 & -12 & 30 \\ -10 & -50 & 20 \end{vmatrix} \Rightarrow \begin{vmatrix} 9 & -18 & 3 \\ -1 & -12 & 3 \\ -10 & -50 & 2 \end{vmatrix} \cdot 10$$

$Det = 9\cdot(-12)\cdot 2 + (-18)\cdot 3\cdot(-10) + 30(-1)\cdot(-50) - ((-10)\cdot(-12)\cdot 3 + (-50)\cdot 3\cdot 9 + 2\cdot(-1)\cdot(-18))\cdot 10$

$Det = -216 + 540 + 150 - (360 + 1350 + 36) \cdot 10$

$Det = 474 + 954 \cdot 10$

$Det = 1428 \cdot 10$

$Det = 14280$

e) $det(AC^t)$, sabendo que $det(AC^t) = det\, A \cdot -det\, C$

$det(AC^t) = Det\, A \cdot - Det\, C \times 1,3$

$$66 \cdot - \begin{vmatrix} 2 & 3 & -1 \\ 6 & 9 & -2 \\ 8 & 12 & -3 \end{vmatrix}$$

$66 \cdot -54 - 48 - 72 - (-72 - 48 - 54)$

$66 \qquad 0$

$Det(AC^t) = 0$

3- Sabendo que $det(A) = -2$

a) $Det(A^t) = -Det(A)$

$Det(A^t) = -(-2)$

$Det(A^t) = 2$

b) $Det(6 \cdot A) = 6^4 \cdot -2$

$Det(6A) = 1296 \cdot (-2)$

$Det(6A) = -2592$

c) $Det(A^7) = (Det\, A)^7$

$Det(A^7) = (-2)^7$

$Det(A^7) = -128$

c) $Det(A^{-1}) = (Det(A))^{-1}$

$Det(A^{-1}) = -\frac{1}{2}$

4- a) $\begin{vmatrix} a & b & c \\ d & e & f \\ 4g & 4h & 4i \end{vmatrix} \xrightarrow{} \times 4 = 4 \cdot -3 = -12$

b) $\times 3 \begin{vmatrix} a & b & -2c \\ 3d & 3e & -6f \\ g & h & -2i \end{vmatrix} \times -2 = -3 \cdot 3 \cdot -2 = 18$

c) $\begin{vmatrix} -a & -b & -c \\ g & h & i \\ -d & -e & -f \end{vmatrix} \begin{matrix} \times -1 \\ \\ \times -1 \end{matrix}$ $\quad l_3 \to l_2$ $= 3 \cdot -1 \cdot -1$

d) $\begin{vmatrix} g & h & i \\ a & b & c \\ d & e & f \end{vmatrix}$ "2 trocar de linhas" $\rightsquigarrow = -3$

e) $\begin{vmatrix} a & b & c \\ 2d+a & 2e+b & 2f+c \\ g & h & i \end{vmatrix} = 2 \cdot (-3) \cdot -6$

f) $\begin{vmatrix} ka+a & kb+b & ck+c \\ d & e & f \\ g & h & i \end{vmatrix}$ $(k+1)a \quad (k+1)b \quad (k+1)c$

$\det = -3 \cdot k+1 \to -3k-3$

5. $A = \begin{vmatrix} 10 & 8 & 40 & -2 \\ 4 & 6 & 20 & -4 \\ -5 & -7 & -30 & 1 \\ 6 & 6 & -30 & 12 \end{vmatrix}$ ($\div 10$ na 3ª coluna) $\Rightarrow \begin{vmatrix} 10 & 8 & 4 & -2 \\ 4 & 6 & 2 & -4 \\ -5 & -7 & -3 & 1 \\ 6 & 6 & -3 & 12 \end{vmatrix} \begin{matrix} \to \div 2 \\ \cdot 10 \\ \to \div 2 \\ \to \div 3 \end{matrix}$

$= \begin{vmatrix} 5 & 4 & 2 & -1 \\ 2 & 3 & 1 & -2 \\ -5 & -7 & -3 & 1 \\ 2 & 2 & -1 & 4 \end{vmatrix} \cdot 10 \cdot 2 \cdot 2 \cdot 3 \Rightarrow$ outra folha continuação:

5- $\begin{vmatrix} 5 & 4 & 2 & -1 \\ 2 & 3 & 1 & -2 \\ -5 & -7 & -3 & 1 \\ 2 & 2 & -1 & 4 \end{vmatrix} \cdot 120$

$2\cdot(-1)^{1+3}\cdot\begin{vmatrix} 2 & 3 & -2 \\ -5 & -7 & 1 \\ 2 & 2 & 4 \end{vmatrix} + 1\cdot(-1)^{3+2}\cdot\begin{vmatrix} 5 & 4 & -1 \\ -5 & -7 & 1 \\ 2 & 2 & 4 \end{vmatrix} + (-3)\cdot(-1)^{3+3}\cdot\begin{vmatrix} 5 & 4 & -1 \\ 2 & 3 & -2 \\ 2 & 2 & 4 \end{vmatrix} + (-1)\cdot(-1)^{3+4}\begin{vmatrix} 5 & 4 & -1 \\ 2 & 3 & -2 \\ -5 & -7 & 1 \end{vmatrix}$

$2(-56+20+6-28-4+60)-1\cdot(-140+8+10-14-40+80)-3(60-4-16+6+20-32)+(15+40+14-15-70-8)$

$2\cdot(-2)-1\cdot(-66)-3(34)+1(-24)$

$(-4)+66-102-24$

$66-130$

$-64 \cdot 120$

-7680

6-a) $\begin{vmatrix} 4 & 6 & x \\ 7 & 4 & 2x \\ 5 & 2 & -x \end{vmatrix} = -128 \Rightarrow \begin{vmatrix} 4 & 3 & x \\ 7 & 2 & 2x \\ 5 & 1 & -x \end{vmatrix}\begin{matrix} 4 & 3 \\ 7 & 2 \\ 5 & 1 \end{matrix} = -64$

$\div 2$

$-8x+30x+7x-10x-8x+21x$

$58x-26x$

$32x=-64$

$x=-2$

b) $\begin{vmatrix} 3 & 5 & 7 \\ 2x & x & 3x \\ 4 & 6 & 7 \end{vmatrix}\begin{matrix} 3 & 5 \\ 2x & x \\ 4 & 6 \end{matrix} = 39$

$21x+60x+84x-70x-54x-28x=39$

$165x-152x=39$

$13x=39$

$x=3$

c) $\begin{vmatrix} x+3 & x+1 & x+4 \\ 4 & 5 & 3 \\ 9 & 10 & 7 \end{vmatrix} \begin{matrix} x+3 & x+1 \\ 4 & 5 \\ 9 & 10 \end{matrix} = -7$

$35x + 105 + 27x + 27 + 40x + 160 - 45x - 180 - 30x - 90 - 28x - 28$

$102x + 292 - 103x - 298 = -7$

$-x - 6 = -7$

$-x = -1$

$x = 1$

d) $\begin{pmatrix} x & x+2 \\ 1 & x \end{pmatrix}$ é singular

$\begin{vmatrix} x & x+2 \\ 1 & x \end{vmatrix} = 0 \Rightarrow x^2 - x - 2 = 0$

$\Delta = 1^2 - 4 \cdot -2 \cdot 1 = 9$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2 \cdot a} \Rightarrow \dfrac{1 \pm 3}{2}$ $\begin{cases} \dfrac{1+3}{2} = \dfrac{4}{2} = 2 \\ \dfrac{1-3}{2} = \dfrac{-2}{2} = -1 \end{cases}$

$\therefore S = (-1, 2)$

e) $\begin{pmatrix} x-4 & 0 & 3 \\ 2 & 0 & x-9 \\ 0 & 3 & 0 \end{pmatrix}$ é inversível $\rightarrow$ Det $\neq 0$

$\begin{vmatrix} x-4 & 0 & 3 \\ 2 & 0 & x-9 \\ 0 & 3 & 0 \end{vmatrix} \begin{matrix} x+4 & 0 \\ 2 & 0 \\ 0 & 3 \end{matrix}$

$0 + 0 + 18 - (0 + (3x - 27) \cdot (x-4) + 0)$

$18 - (3x^2 - 12x - 27x + 108$

$-3x^2 + 39x - 90 \neq 0 \quad \div -3$

$x^2 - 13x + 30 \neq 0$

$\Delta = 13^2 - 4 \cdot 30 \cdot 1$

$\Delta = 169 - 120$

$\Delta = 49$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2 \cdot a} \rightarrow \dfrac{13 \pm \sqrt{49}}{2}$ $\begin{cases} \dfrac{13+7}{2} = \dfrac{20}{2} = 10 \\ \dfrac{13-7}{2} = \dfrac{6}{2} = 3 \end{cases}$

$S = \{x \in \mathbb{R} / x \neq 10 \text{ ou } x \neq 3\}$

7- $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

a) $A^{-1} \rightarrow A A^{-1} = A^{-1} A = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A^{-1} = \frac{1}{det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

A^{-1} existe somente se $det(A) \neq 0$, então

$ad - bc \neq 0$

b) $A = \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} \longrightarrow \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$

$6 - 5 = det(a) = 1$

$B = \begin{vmatrix} 4 & 7 \\ 1 & 2 \end{vmatrix} \longrightarrow \begin{pmatrix} 2 & -7 \\ -1 & 4 \end{pmatrix}$

$8 - 7 = 1 \rightarrow Det(B)$

$(AB)^{-1} = B^{-1} \cdot A^{-1}$

$\begin{pmatrix} 2 & -7 \\ -1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 \cdot 2 + (-7) \cdot (-5) & 2 \cdot (-1) + (-7) \cdot 3 \\ -1 \cdot 2 + 4 \cdot (-5) & -1 \cdot (-1) + 4 \cdot 3 \end{pmatrix}$

$= \begin{pmatrix} 4 + 35 & -2 - 21 \\ -2 - 20 & 1 + 12 \end{pmatrix} \Rightarrow (AB)^{-1} = \begin{pmatrix} 39 & -23 \\ -22 & 13 \end{pmatrix}$

✓

8 - a) $A = \begin{pmatrix} 2 & -2 \\ 3 & 1 \end{pmatrix}$ → cofatores:

$C_{11} = (-1)^{1+1} \cdot 1 = 1$

$C_{12} = (-1)^{1+2} \cdot 3 = -3$

$C_{21} = (-1)^{2+1} \cdot -2 = 2$

$C_{22} = (-1)^{2+2} \cdot 2 = 2$

$CA = \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix}$

b) $B \begin{pmatrix} 2 & -2 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$

$C_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$

$C_{12}\ (-1)^{1+2} \cdot \begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix} = 1$

$C_{13}\ (-1)^{1+3} \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = -1$

$C_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -2 & 0 \\ 1 & -1 \end{vmatrix} = -2$

$C_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 2 & 0 \\ 0 & -1 \end{vmatrix} = -2$

$C_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} = -2$

$C_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -2 & 0 \\ 2 & 1 \end{vmatrix} = -2$

$C_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = -2$

$C_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 2 & -2 \\ 1 & 2 \end{vmatrix} = 6$

$CB = \begin{pmatrix} -3 & 1 & -1 \\ -2 & -2 & -2 \\ -2 & -2 & 6 \end{pmatrix}$

9- Sabendo que: $A^{-1} = \frac{1}{\det(A)} \cdot \operatorname{adj}(A)$

a) $A = \begin{pmatrix} 2 & -2 \\ 3 & 1 \end{pmatrix}$ Det: $2 \cdot 1 + (-3 \cdot (-2))$

det: $2 + 6 = 8$

cofatores de A

$C_{11} = (-1)^{1+1} \cdot 1 = 1$ $\quad C_{21} = (-1)^{2+1} \cdot (-2) = 2$

$C_{12} = (-1)^{1+2} \cdot (-3) = 3$ $\quad C_{22} = (-1)^{2+2} \cdot (2) = 2$

$CA = \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix} \longrightarrow \operatorname{adj}(A) = \begin{pmatrix} 1 & 2 \\ -3 & 2 \end{pmatrix}$

$A^{-1} = \begin{pmatrix} 1/8 & 2/8 \\ -3/8 & 2/8 \end{pmatrix} = \begin{pmatrix} 1/8 & 1/4 \\ -3/8 & 1/4 \end{pmatrix}$

b) $B = \begin{pmatrix} 2 & -2 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix} \Rightarrow \det(B) = -4 + 0 + 0 - 0 - 2 - 2$

$\det(B) = -8$

considerando CB do ex 8, temos

$CB = \begin{pmatrix} -3 & -1 & 1 \\ 2 & -2 & -2 \\ 1 & -2 & 2 \end{pmatrix} \longrightarrow \underset{\downarrow\, CB^t}{\mathrm{adj}}(B) = \begin{pmatrix} -3 & 2 & 1 \\ -1 & -2 & -2 \\ 1 & -2 & 2 \end{pmatrix}$

$B^{-1} = \begin{pmatrix} -3 & 2 & 1 \\ -1 & -2 & -2 \\ 1 & -2 & 2 \end{pmatrix} \div -8 \Rightarrow \begin{pmatrix} 3/8 & -1/4 & -1/8 \\ 1/8 & 1/4 & 1/4 \\ -1/8 & 1/4 & -1/4 \end{pmatrix}$

c) $C = \begin{pmatrix} 0 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} \Rightarrow \det(C) = 0 + 2 + 1 - (0 + 0 + 0)$

$\det(C) = 3$

$C_{11}\ 1 \cdot \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = 1 \quad C_{21} = -1 \cdot \begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} = -1 \quad C_{31} = 1 \cdot \begin{vmatrix} -1 & 1 \\ 0 & -1 \end{vmatrix} = 1$

$C_{12}\ -1 \cdot \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} = -1 \quad C_{22} = 1 \cdot \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 \quad C_{32} = -1 \cdot \begin{vmatrix} 0 & 1 \\ 2 & -1 \end{vmatrix} = 2$

$C_{13}\ 1 \cdot \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2 \quad C_{23} = -1 \cdot \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} = -1 \quad C_{33} = 1\ \begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} = 2$

$Cc = \begin{pmatrix} 1 & -1 & 2 \\ -1 & -1 & 1 \\ 1 & -2 & -2 \end{pmatrix} \longrightarrow \underset{\downarrow\, Cc^t}{\mathrm{adj}\,C} = \begin{pmatrix} 1 & -1 & 1 \\ -1 & -1 & -2 \\ 2 & 1 & -2 \end{pmatrix}$

$C^{-1} = \begin{pmatrix} 1/3 & -1/3 & 1/3 \\ -1/3 & -1/3 & -2/3 \\ 2/3 & 1/3 & -2/3 \end{pmatrix}$

d)

$$D = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 2 & 3 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad det(D) = 0 \cdot | \;\; | - 1 \cdot \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 3 \\ 0 & -1 & 0 \end{vmatrix} + 0 - 0$$

$det \cdot = -1 \cdot -2 - (-3)$

$det = 1$

$C_{11} = 3$	$C_{21} = 2$	$C_{31} = 1$	$C_{41} = 0$
$C_{12} = 2$	$C_{22} = 1$	$C_{32} = 0$	$C_{42} = 0$
$C_{13} = 0$	$C_{23} = 0$	$C_{33} = -1$	$C_{43} = -3$
$C_{14} = 2$	$C_{24} = 0$	$C_{34} = 1$	$C_{44} = 2$

$$C_D = \begin{pmatrix} 3 & 2 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & -1 & 1 \\ 0 & 0 & -3 & 2 \end{pmatrix} \longrightarrow adj\, D = \begin{pmatrix} 3 & 2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & -1 & -3 \\ 2 & 0 & 1 & 2 \end{pmatrix} \Rightarrow D'$$

10- $2A^t = C - XB$

a) $2A^t = C - XB$

$2A^t + XB = C$

$XB = C - 2A^t$

$X = \dfrac{C - 2A^t}{B}$

$X = (C - 2A^t)\, B^{-1}$, onde $det(B) \neq 0$

b) $X = (C - 2A^t) \cdot B^{-1}$

$$A^t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 2 \end{pmatrix} \Big/ \quad B^{-1} : \begin{vmatrix} 3 & -2 & 6 \\ 2 & -1 & 5 \\ 1 & 0 & 3 \end{vmatrix} \longrightarrow B^{-1} = \frac{1}{-1} \cdot adj\, B = -adj\, B$$

$6 + 12 \quad -9 - 10 = -1$

$$C_B = \begin{pmatrix} -3 & -1 & 1 \\ 6 & 3 & -2 \\ -4 & -3 & 1 \end{pmatrix} \Leftarrow$$

$C_{11}(-1)^{1+1} \begin{vmatrix} -1 & 5 \\ 0 & 3 \end{vmatrix} = -3 \quad C_{21} = (-1)^{2+1} \begin{vmatrix} -2 & 6 \\ 0 & 3 \end{vmatrix} = 6 \quad C_{31} = (-1)^{3+1} \begin{vmatrix} -2 & 6 \\ -1 & 5 \end{vmatrix} = -4$

$C_{12}(-1)^{1+2} \begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} = -1 \quad C_{22} = (-1)^{2+2} \begin{vmatrix} 3 & 6 \\ 1 & 3 \end{vmatrix} = 3 \quad C_{32} = (-1)^{3+2} \begin{vmatrix} 3 & 6 \\ 2 & 5 \end{vmatrix} = -3$

$C_{13}(-1)^{1+3} \begin{vmatrix} -2 & 6 \\ 0 & 3 \end{vmatrix} = 1 \quad C_{23} = (-1)^{2+3} \begin{vmatrix} 3 & -2 \\ 1 & 0 \end{vmatrix} = -2 \quad C_{33} = (-1)^{3+3} \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = 1$

Sabendo que:

$$A^t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 2 \end{pmatrix} \qquad \operatorname{adj}(B) = \begin{pmatrix} -3 & 6 & -4 \\ -1 & 3 & -3 \\ 1 & -2 & 1 \end{pmatrix} \Rightarrow B^{-1} = \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$B^{-1} = -\operatorname{adj}$

$$X = (C - 2A^t) B^{-1}$$

$$X = \left(\begin{pmatrix} 4 & 4 & 0 \\ 0 & 8 & -2 \\ -2 & 0 & 6 \end{pmatrix} - 2 \cdot \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$$X = \left(\begin{pmatrix} 4 & 4 & 0 \\ 0 & 8 & -2 \\ -2 & 0 & 6 \end{pmatrix} - \begin{pmatrix} 2 & 4 & 0 \\ 0 & 6 & -2 \\ -2 & 0 & 4 \end{pmatrix} \right) \cdot \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$$X = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$$X = \begin{pmatrix} 2 \cdot 3 + 0 \cdot 1 + 0(-1) & 2 \cdot (-6) + 0 \cdot (-3) + 0 \cdot 2 & 2 \cdot 4 + 0 \cdot 3 + 0 \cdot (-1) \\ 0 \cdot 3 + 2 \cdot 1 + 0 \cdot (-1) & 0 \cdot (-6) + 2 \cdot (-3) + 0 \cdot 2 & 0 \cdot 4 + 2 \cdot 3 + 0 \cdot (-1) \\ 0 \cdot 3 + 0 \cdot 1 + 2 \cdot (-1) & 0 \cdot (-6) + 0 \cdot (-3) + 2 \cdot 2 & 0 \cdot 4 + 0 \cdot 3 + 2 \cdot (-1) \end{pmatrix}$$

$$X = \begin{pmatrix} 8 & -12 & 8 \\ 2 & -6 & 6 \\ -2 & 4 & -2 \end{pmatrix}$$